NRES 746 – Week 5, Lecture 2: Writing the Likelihood

Answer key. Answers are in magenta.

Hand this in Monday. The instructor is not here today but make sure your name is at the top and turn it in next Monday!

Instructions: Work alone on Parts 1-3, check in with your group, then work through Part 4 with your group. Please keep laptops closed through Part 4 – calculators/phones are fine for arithmetic, and fractions are fine as final answers. You can use laptops and Rstudio for Part 5. If you get stuck, note where and move on.

Putting it in context

On Monday you went *forward*: you chose a data-generating model and imagined the data it would produce. Today we go *backward*: the data are already in hand (they are the "known", the fixed quantity), and the question is what they tell us about the parameters (which are the "unknown", or variable quantity). The probability formulas are the ones you already know from Bolker Ch 4.

We will stay with the sage-grouse nest-fate example from Monday. Here are four nests, with outcome $y_i = 1$ if at least one egg hatched and $y_i = 0$ if the nest failed:

Nest	Sagebrush cover (%)	Outcome (1 = hatched, 0 = failed)
1	30	1
2	20	1
3	10	0
4	30	1

Part 1: One observation (individual, ~4 min)

For now, suppose every nest has the same unknown probability p of success.

1a. Write down the probability of each possible outcome for a single nest.

$P(y = 1 \mid p) = p$ $\quad$ $P(y = 0 \mid p) = 1 - p$

1b. Write a single formula that gives the probability of an observed outcome y. Make sure the formula gives the correct answer for both possible outcomes of y (0 or 1).

$P(y \mid p) = p^y(1-p)^{1-y}$ $\quad$ (also correct: $yp + (1-y)(1-p)$)

> **Definition.** The **likelihood** of a parameter value is the probability of the data you actually observed, if that parameter value were the truth. It is written as a function of the parameter alone, with the data held fixed – not as a conditional probability of the parameter given the data, even though it is built from one. A subscript marks how much data is held fixed behind the scenes: for a single observation, $\mathcal{L}_1(p) = P(y \mid p)$.

1c. Nest 1 hatched ($y_1 = 1$). Evaluate the likelihood of $p = 0.75$ and of $p = 0.25$ given this observation.

$\mathcal{L}_1(0.75) = 0.75$ $\quad$ $\mathcal{L}_1(0.25) = 0.25$

1d. In your formula from 1b, which symbol is the data and which is the parameter?

y is the data; p is the parameter.

Part 2: Several observations (individual, ~8 min)

Now use all four nests. Assume every nest has the same success probability p, and that the nests are independent of one another.

2a. Write the likelihood of all four observations, $\mathcal{L}_4(p)$, in terms of your formula from 1b. We can assume all observations are independent! Please use product notation ($\prod$) to write the likelihood.

$$\mathcal{L}_4(p) = \prod_{i=1}^{4} p^{y_i}(1-p)^{1-y_i}$$

2b. Plug in the observed outcomes and simplify to the form $p^j(1-p)^k$. What are j and k?

$j = 3$ $\quad k = 1$ $\quad$ (so $\mathcal{L}_4(p) = p^3(1-p)$; $\binom{4}{3}p^3(1-p)$ also accepted)

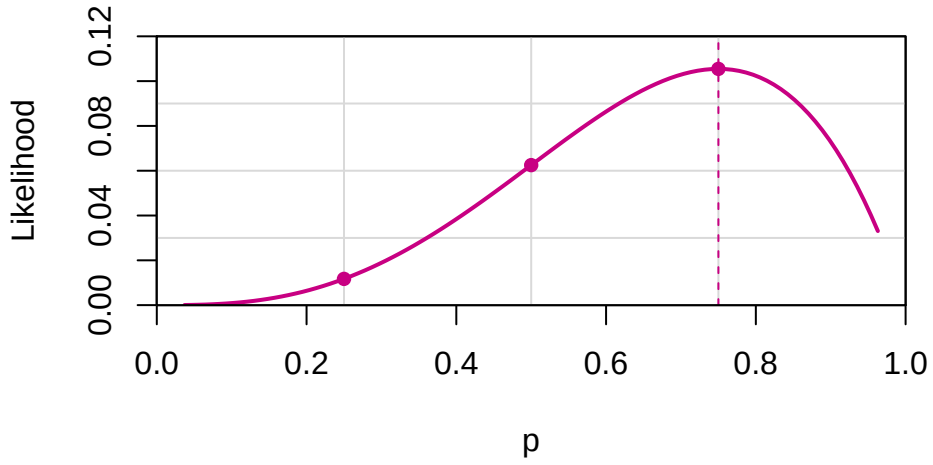
2c. Evaluate your answer to 2b at the three values of p below. Fractions are fine.

p	Likelihood
0.25	3/256 = 0.0117
0.50	1/16 = 0.0625
0.75	27/256 = 0.1055

2d. Add up your three likelihood values. Do they sum to 1? Should they?

$46/256 \approx 0.18$. No, and they shouldn't: a likelihood is not a probability distribution over p.

2e. Plot your three values on the axes below, then sketch the whole curve $\mathcal{L}_4(p)$ for $0 \le p \le 1$. Mark where you think the peak is. Is that where you would have guessed from the raw data alone?



Peak at $p = 0.75$, matching 3 of 4 nests hatched.

Part 3: The log-likelihood (individual, ~3 min)

3a. Take the log of your product from 2a and use the log rules to rewrite it as a sum. Write your result for a general set of n nests with outcomes $y_1, \ldots, y_n$.

$\ell_n(p) = \log \mathcal{L}_n(p) = \sum_{i=1}^{n} [y_i \log p + (1-y_i)\log(1-p)]$

(equivalently $(\sum y_i)\log p + (n - \sum y_i)\log(1-p)$)

3b. Suppose the monitoring program had 1,852 nests and each nest contributed a likelihood of about 0.5. Roughly how big is the product of all 1,852 terms? Note that standard computer arithmetic cannot store positive numbers much smaller than 10^{-308}. Why might you want to work with the sum from 3a instead? What is its value here (feel free to use your phone or a calculator)?

$0.5^{1852} \approx 10^{-558}$, which a computer stores as 0 (underflow). The sum is an ordinary number: $1852 \log(0.5) \approx -1283.7$.

Group check-in (~3 min)

Compare your answers to Parts 1-3 with your group. Where you differ, argue it out and revise. Then answer together:

Group question. Multiplying the four nest likelihoods together depended on an assumption. Name one biological reason it might fail for real sage-grouse nests.

Independence. Any shared cause works, e.g. the same hen renesting, neighboring nests sharing predators, shared weather in a year.

Part 4: Adding the covariate (with your group, ~6 min)

Monday's model had success probability rising with sagebrush cover. A logistic function keeps it between 0 and 1:

$$p_i = \frac{1}{1 + e^{-(a+bx_i)}}$$

where x_i is the cover around nest i. Every nest now has its own p_i. Suppose a candidate pair of parameter values is $a = -2.2$ and $b = 0.11$. Here are the success probabilities it predicts (rounded) for our four nests:

Nest	Sagebrush cover (%)	Outcome (1 = hatched, 0 = failed)	Predicted p
1	30	1	0.75
2	20	1	0.50
3	10	0	0.25
4	30	1	0.75

4a. Check the predicted p for nest 2 (20% cover) by hand.

$-2.2 + 0.11(20) = 0$, so $p_2 = 1/(1 + e^0) = 0.5$.

4b. Assuming the nests are independent, what is the likelihood of this candidate parameter pair, $\mathcal{L}_4(a = -2.2, b = 0.11)$? Careful with the nest that failed.

$0.75 \times 0.5 \times (1 - 0.25) \times 0.75 = 27/128 \approx 0.211$ (unrounded p_i: 0.211).

4c. Compare this to the best of your three constant-p likelihoods from 2c. Which model do the data support more, and by what factor? Does that show that cover really matters for nest success? Give at least one reason for caution.

Covariate model, by a factor of 2 (27/128 vs. 27/256). No: e.g. only four nests; two parameters vs. one; (a, b) wasn't fitted; no measure of uncertainty.

4d. Write the likelihood for n nests in general, $\mathcal{L}_n(a, b)$, using $p_i = 1/(1 + e^{-(a+bx_i)})$. Then underline the quantities that are fixed data and circle the quantities that the likelihood is a function of.

$$\mathcal{L}_n(a, b) = \prod_{i=1}^{n} p_i^{y_i}(1 - p_i)^{1-y_i}, \qquad p_i = \frac{1}{1 + e^{-(a+bx_i)}}$$

Underlined (data): y_i, x_i (and n). Circled: a, b.

Laptops open now. From here on, actually open RStudio and run each step as you go, rather than just working it out on paper. Resist the temptation to hand this one to an AI assistant!

Part 5: From math to R code (with your group, ~10 min)

Later in the course you will fit models like this by writing the likelihood as an R function and handing it to an optimizer. This is your first try, so expect a little trial and error – that's normal.

The built-in `dbinom(x, size = 1, prob = p)` gives the Bernoulli probability from Part 1: `x` is the observed outcome (0 or 1), `size` is the number of trials per observation, and `prob` is the success probability. Setting `log = TRUE` returns the log of the probability.

5a. Type this function into R, filling in the blanks so it returns the *negative* log-likelihood of a vector of nest outcomes `y` for a single success probability `p`. Run it – if R complains, that's useful information, not a failure.

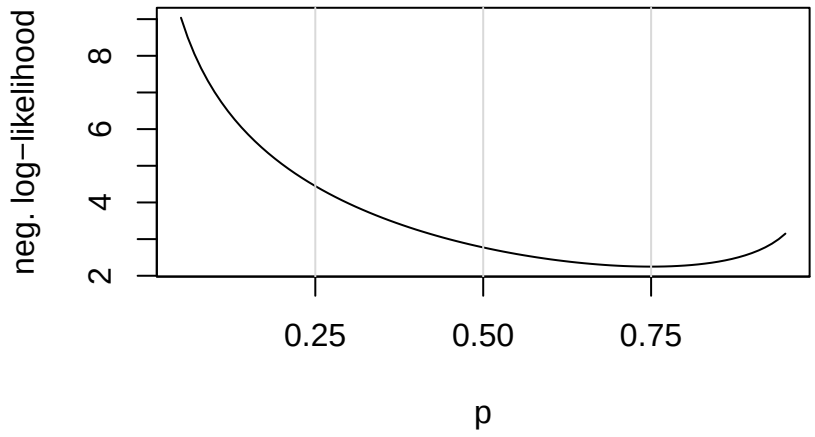
```
nll_const <- function(p, y) {
  -sum(dbinom(x = y, size = 1, prob = p, log = TRUE))
}
```

5b. Run `y <- c(1, 1, 0, 1)`, then `exp(-nll_const(0.5, y))`. Which entry of your table in 2c should it match? If it doesn't, use the mismatch to track down the bug.

0.0625, the $p = 0.50$ row (1/16).

5c. Run the code below with your own working `nll_const`. (`sapply()` evaluates a function at each value of a vector in turn.)

```
p_grid <- seq(0.05, 0.95, by = 0.01)
nll_vals <- sapply(p_grid, function(p) nll_const(p, y))
plot(p_grid, nll_vals, type = "l", xlab = "p",
     ylab = "neg. log-likelihood", xaxt = "n")
axis(1, at = seq(0, 1, 0.25))
abline(v = seq(0, 1, 0.25), col = "grey85")
```



Which value of p gives the smallest negative log-likelihood? Does it agree with your sketch in 2e? Why do you think we work with the *negative* log-likelihood rather than the log-likelihood?

$p = 0.75$; yes. Optimizers minimize by default (like SSR); minimizing $-\ell$ is the same as maximizing ℓ.

Reflection. (individual) Now that you've written and run your first likelihood function, what part of turning the math into code gave you the most trouble? And more generally, what is still fuzzy about likelihood – what would you like the instructor to review on Monday?

Open-ended.

5d. Optional challenge. Write the covariate version, using your formula from 4d.

```
nll_cover <- function(par, y, cover) {
  p <- 1 / (1 + exp(-(par[1] + par[2] * cover)))
  -sum(dbinom(y, size = 1, prob = p, log = TRUE))
}
```

If you ran `exp(-nll_cover(c(-2.2, 0.11), y, cover))`, what number should you get, approximately, and why?

About 0.211: the likelihood from 4b (slightly different because 4b used rounded p_i).